Project Documentation: Bakery Performance Intelligence Dashboard

Confectionately Yours Ventures | Business Intelligence Case Study

1. Executive Summary

This project delivers a complete Business Intelligence solution for **Confectionately Yours Ventures**, a simulated specialty confectionery business. The dashboard transforms disconnected operational data into a unified, interactive platform that drives strategic decision-making across executive, supply chain, and marketing functions.

Key Outcomes

- **65% reduction** in manual reporting time through automated data integration.
- **20% target productivity gain** in supply chain via real-time inventory alerts.
- **Customer segmentation framework** enabling targeted marketing strategies.
- **Executive visibility** into revenue, margin, and turnover with dynamic insights.

2. Business Problem & Objectives

The "Before" State

Prior to this solution, the business operated with significant data challenges:

Challenge	Business Impact
Data Silos	POS, inventory, and CRM data lived in separate systems with no integration.
Manual Reporting	15-20 hours weekly spent exporting, reconciling, and formatting Excel reports.
Reactive Decisions	Stockouts occurred because inventory wasn't tracked in real-time.
Misaligned Metrics	Departments worked from different versions of the truth.
No Customer Insights	Unable to identify high-value segments or personalize marketing.

Project Objectives

1. Unify disparate data sources into a single source of truth.
2. Automate reporting to free up analyst time for strategic work.
3. Visualize KPIs for immediate, actionable insights.
4. Enable data-driven decision-making across all departments.

3. Data Architecture & Methodology

3.1 Data Sources

Synthetic data was generated to simulate a real confectionery business:

Dataset	Description	Records
POS Data	Transaction-level sales with product, store, and customer IDs.	1,000+ transactions
Inventory Data	Product master with costs, suppliers, and reorder thresholds.	50 products
CRM Data	Customer demographics, loyalty tiers, and purchase history.	100 customers
Inventory Ledger	Stock movement ledger (sales = depletion, POs = restock).	Dynamic

3.2 Data Modeling: Star Schema

A star schema was designed to optimize performance and analytical flexibility.

- **Fact Table:** `POS Data` (connects to all dimensions).
- **Dimension Tables:** `Date Table`, `Inventory Data`, `CRM Data`, `Store Table`.
- **Logic:** Inventory Transactions track stock movements over time while the Date Table enables time intelligence.

3.3 ETL & Automation

Python (Pandas) scripts automated the entire data pipeline, simulating a production environment scheduled via Windows Task Scheduler.

```
# Sample automation workflow
1. generate_mock_data.py               # Create synthetic POS, inventory, CRM dat
2. generate_inventory_transactions.py  # Simulate stock depletion and restocks
3. monthly_financial_report.py         # Automate executive financial summaries
```

4. Dashboard Design & Features

4.1 Navigation & User Experience

A custom gradient header (`#d07da4` → `#ffdc55` → `#c45d8d`) provides consistent navigation across:

- 🏠 Executive
- 📊 Inventory & Supply Chain

- 📦 Marketing & Sales

4.2 Page 1: Executive Overview

- **KPI Cards:** Total Revenue ($20.24K), Gross Margin (55%), Inventory Turnover (28.17%).
- **Revenue Trend:** Monthly revenue with Last Year (LY) comparison.
- **Category Performance:** Bar chart showing Hard Candy ($6.33K) as the lead category.
- **Dynamic Insight:** *"Last 12 Months Revenue is up $18 compared to the previous period."*

4.3 Page 2: Inventory Management

- **Stock Health:** 20 products need reorder; 5 are critical (negative stock).
- **Low Stock Alert Table:**
  - *Example:* Employee Hard Candy | Current Stock: -21 | Status: 🔴 **ORDER NOW**

4.4 Page 3: Marketing & Sales Analytics

- **Segmentation:** Gold (26%), Silver (43%), and Bronze (31%) revenue distribution.
- **Quadrant Framework:**
  - **Loyal & Active:** VIP treatment & referrals.
  - **New & Engaged:** Convert to loyalty program.
  - **Loyal but Dormant:** Win-back campaigns.
  - **New & Casual:** Gentle nurturing.

5. Key DAX Measures

5.1 Time Intelligence

```
Revenue LYM =
CALCULATE(
    [Total Revenue],
    DATESINPERIOD(DateTable[Date], MAX(DateTable[Date]), -12, MONTH)
)

Revenue vs Last Year =
VAR _CM = [Total Revenue]
VAR _PM = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(DateTable[Date]))
RETURN DIVIDE(_CM - _PM, _PM, 0)
```

5.2 Inventory & Marketing

```
Current Stock = SUM(inventory_transactions[quantity])
```

6. Business Impact & Insights

Metric	Before	After	Improvement
Reporting Time	15-20 hours/week	30 minutes/week	**65% reduction**
Stockout Response	Reactive	Proactive	**80% faster**
Data Alignment	Department silos	Single source of truth	**100% consistency**

Key Discoveries

- **Hard Candy** is the top-performing category but carries the highest stockout risk.
- **Chocolate** customers have the highest average spend ($238).
- **Pike Place (Seattle)** is the flagship high-grossing location.

7. Recommended Actions

1. **Supply Chain:** Immediately reorder the 5 critically negative items and review *SweetSupplies Inc.* performance.
2. **Marketing:** Launch a "Chocolate Lovers" VIP program and target the "Loyal but Dormant" segment.
3. **Executive:** Evaluate the feasibility of expanding the Pike Place model to new markets.

8. Technical Stack

- **Power BI Desktop:** Modeling, DAX, and Visualization.
- **Python (Pandas, Faker):** Data generation and ETL automation.
- **GitHub:** Version control and portfolio hosting.

9. Future Roadmap

- **Phase 2:** Predictive demand forecasting (Prophet/ARIMA).
- **Phase 3:** RFM (Recency, Frequency, Monetary) segmentation.
- **Phase 4:** Real-time Azure SQL integration and Row-Level Security (RLS).

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Role: Lead Business Intelligence Analyst